

EbCRD – Final assignment

As announced in class: the final project is a small Unity game that you create on your own, and a video that you create to showcase that game and explain the most important parts. Please read the following document carefully.

- Please upload your video directly to the **eCampus** under **UPLOAD 1**: either the video file or a ZIP containing the video, or a .txt file including a link to an online portal where we can watch the video (e.g., YouTube), or a .txt file with a download link to the video - in that case, make sure we can access the file and that it is not only temporarily reachable (as it would be with services like WeTransfer).
- The total length of the video should be between 15 and 25 minutes.
- The video should have two parts: in the first part, you show the gameplay and the mechanics. This should not take longer than **a maximum of 7 minutes**. Show the main feature(s) and how the game works. In the second part, you should explain the project and the code. Details follow below.
- In the second part of the video, please explain how you implemented the most important parts of the game in Unity.
- **Deadline:** As announced in class, you can upload your project and video either by the end of the first week of July (**6th of July**) or by the beginning of the next semester (**21st of September**). **Both count as a first attempt**. There is no difference in which timeslot you choose, except that you will get your grade already in August if you go for the first deadline.
- **If you fail on the first attempt, there will be a second attempt available until the end of October.**
- The goal is **to create** a small game with one or two core functions.
- The total time you should invest should not exceed 50 hours (including the video). Ideally, it should be around 40 hours. So don't invest too much time in "huge game design." This is not comparable to the CCL. The main purpose of this assignment is to show that you can handle basic tasks in Unity. Even if not everything works perfectly in the end – what counts most here is the project setup and, most of all: that you are able to explain it clearly.
- As a reference: The small game you did in class and for Lucas's non mandatory was approximately 20–25 hours in total.
- This assignment should ensure that everyone has a basic understanding of Unity essentials (since not everyone in the CCL worked with Unity).

- Partially comparable to what we did in class: you create a game with a few scripts that cover a few mechanics.
- The quality of the video is not graded at all. Just make sure your screen is recorded and that your voice is clearly understandable (**important!!!**).
To ensure this works, make a short test recording just before the actual recording for the assignment, and then play that test video file to check if you are clearly understandable. This helps to confirm that all settings are good. Then start with the actual recording.
- We recommend using OBS (Open Broadcaster Software | OBS) for screen recordings, but you can use whatever you like. <https://obsproject.com/>
- You don't need to cut anything. If you explain something incorrectly and correct yourself afterward, that's totally fine. Also, if you have gaps in your explanation because, e.g., you have to look something up – no problem at all.
- Still: try to have a rough plan for how you're going to explain the project before you start recording. This should make it easier for you to do the recording.
- In the end, the video should give us a brief impression of the project you've implemented, its features, and how it is played in the first part.
- We do not grade any 3D models in this course. So, this **does not have** to be done on your own - feel free to use graphical assets from the Asset Store or other sources. Of course, feel free to use stuff you have created on your own – but it's not part of the grade for this assignment and course. Or use Unity primitives. But to some degree, aesthetics are part of the grade – not extremely, but if you only have a few primitives and no aesthetics at all, you will lose a few points. More about the grading criteria is found later in the document.
- Audio implementation is not needed. You might get some extra points for additional aesthetics if you include sound here and there, but it's not a requirement. Feel free to use WWise or Unity built-in tools for sound. Of course, feel free to use sounds you have created on your own – but again, it's not part of the grade for this assignment and course.
- On the other hand, code that you have written on your own is graded – but you don't have to explain every line (!!). Just the most important/complex/interesting parts need to be explained.
- Also, complex lighting setups or the use of complex third-party assets will be graded. So, not only code. However, be sure that you can explain how and why these things work as they do in your game. It doesn't count if you say, "because I've seen that in tutorial XYZ" or "ChatGPT told me to do so."

- The same applies to complex animation configurations, input system configurations, XR usage, etc. – everything that you have done on your own in Unity that has a certain complexity (so not only moving something from A to B and back), **as long as** you can explain it (!).
- Even if parts of the project are based on a tutorial, it's still fine if you can explain how it works. So, this video should act as a “proof” that you understand your own project.
- The better you can explain your project and show that you understand how the various parts of your game work (and work together), the more points you'll get.
- If your video gives the impression that you've just followed steps without really understanding the core mechanics of your project, you'll get fewer points for the respective part.
- Be aware: the main source of your grade will be this video. However, if we want to look something up in your project (because we have questions after watching the video), we might also check the project for clarification. So please upload your project under **UPLOAD 2**: either a ZIP file containing the project **or** a .txt file including a link to an online portal with a download link to the ZIP file – in that case, make sure that we can access the file and that it is not only temporarily reachable (as with WeTransfer).
In any case: delete the Library folder before you ZIP the project.
- The maximum score you can get for this assignment is 65 + 15 extra bonus points for additional features (**details follow below**).
- For Yun's assignment, you got up to 15 points.
- For Lucas's assignment, you got up to 15 points as well.
- You got 5 points if you took part in Yun's VR class.
- That means, by now you can have 35 points (Yun's assignment, Lucas's assignment, VR class). And you can get 65 additional points for this assignment through regular means.
Plus: up to 15 extra points for the additional features listed below. These optional features are especially an offer for those who didn't upload the (non-mandatory) assignment for Lucas's class. Of course, anyone can aim for these additional 15 points.
- This means, in theory, you can get up to 115 points for the course in total.
- However, points are not given easily 😊. So, we are critical when it comes to grading your project. Here are the main criteria that will be looked at:

- **Top Priority:**
 - o Explanation
- **High Priority**
 - o Game features/mechanics
 - o Implementation in Unity (Unity Features/Inspector/Prefabs/General Game Setup)
 - o Implementation in C#
 - o Structure (Code/Folders/Scene Hierarchy)

These areas (Top & High priority) make up around two-thirds of the 65 points you can get.

- **Also important for grading (the other third)**
 - o Originality
 - o Complexity
 - o Aesthetics
 - o Unity Features
- About the time you should invest: really try to cut off the project around 35–45 hours, and then create the video (unless, of course, you *want* to invest more time). The whole course EbCRD has 5 ECTS. This equals approximately 125 hours of work. You had 45 × 45 minutes in class. That sums up to ~34 hours.
We estimate that you invested around 30 additional hours on the assignments from Yun and Lucas. And we estimate that you will need around 40 hours for the final assignment. This leaves about 20 hours of buffer time, which you might have partly used to deepen your Unity knowledge in preparation for the CCL, or which you can use if some assignments take you longer than expected.

You can even use your CCL project as a starting point for this assignment. However, the features you're implementing additionally must be very clearly explained in the video. **It's not enough to just add "new levels with the existing mechanics."**

As mentioned earlier, you can get up to 15 extra points:

- **Five points** if your assignment contains some form of Procedural Generation (details in the box)

- *Eg. Use the C# final assignment algorithms, such as Kruskal, to generate a map;*
- *OR use some other technique from the internet or seen in class for generating items, levels, etc*
- *OR Part of the aesthetics, for example with Shader Graph or coded shaders.*
- ***If you go for the latter: What is important is that this is somehow connected to your game architecture in such a way that you are controlling the procedural generation through a value/multiple values in the code (not just a static shader).***

- **Another 10 points** if you do a VR project.